

Code for: Managers and Productivity in Retail

Robert D. Metcalfe, Alexandre B. Sollaci, and Chad Syverson
Journal of Political Economy

Overview

The replication folder contains 3 sub-folders that include all of the required data and codes to replicate the paper: **Code**, **Data**, and **Results**. The Code folder contains 3 sub-folders, **R Codes**, **Stata Codes**, and **NAM Simulations**. The first two contain the scripts in R and Stata, respectively, and the last has the scripts used for the simulating the NAM selection results (in Stata and Python). The Results folder also contains two sub-folders, **Figures** and **Tables**. In addition to those, there is an **ado** folder that holds all required Stata packages, and a **logs** folder that saves the output from Stata's command window. The directory organization is sketched below.

```
Code_for_Managers_and_Productivity_in_Retail/  
├── ado/  
├── Code/  
│   ├── NAM Simulations/  
│   ├── R codes/  
│   └── Stata codes/  
├── Data/  
├── logs/  
└── Results/  
    ├── Figures/  
    └── Tables/
```

The code can be run through two main files, located in the root folder: **01_main.R** and **02_main.do**. They should be run in this order, as the 01_main.R estimates the fixed effects that are later used by 02_main.do. The python scripts are called from within 02_main.do, but if the replicator prefers they can be run independently in a Python compiler (no particular order required).

The full replication should run for about one hour and ten minutes (approximately 1 hour for 01_main.R and 10 minutes for 02_main.do).

Data Availability

Data from the two retailers was provided under the condition that companies remain anonymous. As such, the data used for this paper cannot be made publicly available. The JPE Data Editor has conducted plausibility checks concerning data provenance under an NDA with the authors.

The authors commit to preserving the data and code for a period of at least five years following the publication of the paper and to provide reasonable assistance to requests for clarification and replication.

Statement of Rights

We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Dataset List

The main data sources for the analysis are the data provided by each company. They are saved on the "Data" folder as Stata dta files: "A__dataset.dta" and "B__dataset.dta". A full data dictionary with the variables included in each of these datasets is provided at the end of this document in Tables 2 and 3 at the end of this document. Several other datasets are created for the remainder of the analysis. These are listed and described in the Table 1 below.

Computational Requirements

There are no specific hardware requirements to run the codes provided, and most modern laptops should be able to complete them in about 1 hour. Codes were run on a Lenovo ThinkPad T14s laptop, with 32GB of installed RAM and a 2.00 GHz processor (AMD Ryzen AI 7 PRO 350 w/ Radeon 860M). The operating system was Windows 11 Enterprise. The replicator should be able to run scripts on Stata, R, and Python. The specific libraries and dependencies required in each software are listed below.

- R (version 4.4.0) will require the following packages to be installed:
 - haven
 - lfe
 - doBy
 - RhpcBLASctl
 - Matrix (this is not explicitly called as a library, but is required for ‘lfe’ to work properly)
- Stata (version 19) will require the following packages to be installed (all available through ssc):
 - reghdfe
 - ftools
 - require
 - estout
 - did_imputation
 - event_plot
 - binscatter
 - grc1leg2
 - white_tableau scheme (can be installed through: ssc install schemepack)
- Python (version 3.11.11) will require the following libraries:
 - numpy
 - pandas
 - scipy
 - matplotlib

In addition to those, replication will also require the **ebayes.ado** and **fese_fast.ado** programs written by Adam Sacarny (see <https://sacarny.com/programs/>). For convenience, these can be found within the **Stata codes** sub-folder within the **Code** folder.

Instructions to Replicators

The root directory contains two main files, **01_main.R** and **02_main.do**, which will run the entire replication of the paper. To successfully replicate the paper, the user should

1. Run the **01_main.R** in RStudio
 - Other IDEs and compilers should work as well, but the definition of working directories is done through the **rstudioapi** library
2. Run the **02_main.do** in Stata
 - For this file to run properly, the user will have to correctly set the path in "global root" macro:


```
// Set path to replication file
global root "[correct path]/Replication_for_Managers_and_Productivity_in_Retail"
```
 - This file automatically installs all of the required packages to run the replication.
 - It also briefly describes what each of the scripts it calls will do, and the tables/figures it produces.
 - Lastly, the 02_main.do also calls and runs the Python scripts in its last section. However, these scripts do not take any of the global variables from Stata, and can thus be run independently if desired.

Navigating the code files: The order of the results in the following code does not always follow the order in which they are presented in the paper. To facilitate the navigation between code and results, the 02_main.do file indicates which results are produced in each do file it calls. These will appear as a **commented code** right before the "do [file name]" command.

For example, the code excerpt below indicates that the file "A_prelim_analysis.do" produces Figures A.1 and C.1, and Tables B.1, C.1, and D.1.

```
* Preliminary look at the data: Produces Figures A.1 and C.1; Tables B.1, C.1, and D.1
do "$code_dir/A_prelim_analysis.do"
```

Naming conventions: All figures are saved in the Results/Figures folder and all tables are saved in the Results/Tables folder. The naming should be easy to follow, as it matches their placement in the paper. For example, Figure 3, panel (a) will be saved as "Figure_3_panel_A", and similarly for other figures.

In the case of the event study figures, they are named based on the their outcome variable: e.g., "Figure_6_sales" refers to Figure 6, panel (b), which shows the impact of managerial moves on sales. The mapping between figure names and outcomes is:

Outcome variable	Figure name (suffix)
Productivity	<code>__prod</code>
Sales	<code>__sales</code>
Full-time Employment	<code>__ft</code>
Part-time Employment	<code>__pt</code>
Full/Part-Time Employment	<code>__ft_ratio</code>
Manager Salary	<code>__mng_salary</code>
Energy Consumption	<code>__energy</code>

Description of the Code Scripts

We provide a detailed description of all scripts and the results they provide below. As before, we use X as a placeholder for A or B in the names of scripts that are used for both companies A and B.

- **X_estimate_fe.R** provides the initial estimates of the manager, store, and time fixed effects from equation 1 in the paper. It also adjusts the variance of the fixed effects for limited mobility bias and computes the variance/covariance shares in the four largest connected sets, as described in [Table 1](#).

Constructing [Table 1](#): This script will save the variance/covariance shares in the files "j-X_cs_covar.txt", for $j = 1, \dots, 4$ representing the four largest connected sets (for example, "2-A_cs_covar.txt" has the variance/covariance shares for the second largest connected set in Company A). These files can be found in the "Results/Tables" folder. The number of observations within each connected set shown in [Table 1](#) is obtained after merging the "X_store_fe.dta" and "X_manager_fe.dta" datasets back into the original "X_dataset.dta", and is done in the "X_analyze_FE.do" file (it will be displayed in the Stata window when run).

- **X_estimate_gfe.R** clusters all managers and stores into groups based on observables and estimates the fixed effect of each group using equation 1. As above, it also adjusts the variance and covariance of estimates for limited mobility bias and saves them in the file "X_covar_group.txt" (in this case there is a single connected set). This file is again found in the "Results/Tables" folder.
- **X_analyze_FE.do** takes the initial estimates of the fixed effects (above) and adjusts them using the Empirical Bayes Adjustment. This is done with the help of the programs **ebayes.ado** and **fese_fast.ado**. Using these results, the script
 - Plots the distribution of fixed effects ([Figure 3](#), [Figure A.2](#), and [Figure A.3](#)) and the number of stores and managers in each connected set ([Figure A.4](#));
 - Runs the out-of-sample test ([Figure 4](#));
 - Performs the series of robustness checks discussed in [Appendix C.2](#). This includes [Figure C.2](#) and [Figure C.3](#), as well as the correlations (paragraph on local shocks) and the R^2 numbers (paragraphs on persistence and learning) in [Appendix C.2](#).
 - Estimates the full covariance matrix for manager and store fixed effects and performs the exercises discussed in the "Correlated Measurement Error" section of [Appendix E.2](#) (see the last paragraph for the calculated ratios).

- **X_analyze_GFE.do** analyzes the grouped fixed effects estimates, producing the Figures in [Appendix E.1](#).
- **X_event_study.do** runs all of the analysis related to the event studies. This includes [Figure 5](#) and [Figure 8](#) for Company A, [Figure 6](#) and [Figure 9](#) company B, and all Figures in [Appendix G](#).

Using the estimates from the event studies, this script also produces [Table 4](#), [Figure 7](#) and [Figure A.5](#). After breaking managers and stores into terciles based on their qualities, the script produces [Figure D.1](#), as well as [Table 2](#) and [Table 3](#).

- **X_prelim_analysis.do** calculates the summary statistics mentioned in the beginning of [section 2](#) in the paper. The script also does some of the preliminary analyzes of the difference between "movers" and "non-movers", producing [Figure A.1](#), [Table B.1](#), and [Table D.1](#). Finally, it compares different measures of productivity and produces the results discussed in [Appendix C.1](#).
- **X_revenue_counterfactuals.do** does all of the analysis reported in [Appendix F](#), including [Table F.1](#).
- **B_gender_moves.do** (only available for company B) runs the regressions required for [Table B.2](#) and calculates the numbers mentioned in [Footnote 7](#).

- **selection_based_NAM.do** is a do file that generates [Figure 2](#) and [Figure D.2](#).
- **simulate_selection.py** is a python script that generalizes the simulations from the "selection_based_NAM.do" file into 20 rounds of (re)matching of managers and stores. This is used to produce the three charts in [Figure D.3](#).
- **calibrate_NAM.py** uses the observed variances and covariances in the data ("NAM_data.xlsx") to recover the size of the bias in the correlation between managers and stores using the expression derived in the "Discussion and Extensions" part of [Section 3.3](#) and further developed in [Appendix D.2.1](#).
- **NAM_data.xlsx** is an excel file that simply calculates the weighted averages of covariances from [Table B.1](#). The numbers found in this file are used in the 'calibrate_NAM.py' program.

Replicating In-text Numbers

Several numbers obtained from the analysis are simply discussed in the paper's text, and not present in any Table or Figure. We indicate where they can be found within the replication package below.

- **p. 8** "about 60% change managers only once, 30% change managers twice, and the remainder more frequently than that."

```
- X_prelim_analysis.do: tab num_managers_store if aux == 1 & num_managers_store > 1
```

Similarly, 14 or 23% of managers (depending on the company) work in two different stores, and a small fraction move through three or more.

```
- X_prelim_analysis.do: tab num_stores_manager if aux == 1
```

Between 30 and 50% of each company's manager changes are internal"

```
- X_prelim_analysis.do: gen share = total_internal/total_changes, followed by su share
```

- **p.10** Footnote 7 "Female managers are about 7% less likely to move to another store than male managers, a gap that cannot be explained by manager tenure or store characteristics"

```
- B_gender_moves.do: bys female: tab mover
```

- **p. 11** "Our data contains about 890 connected sets in Company A and just over 1,200 connected sets for Company B" // "The four largest connected sets in Company A have between 13 and 18 managers and 7 to 10 stores." // **p. 12** "Company B. The largest set has 202 managers and 131 stores." // "Set sizes drop quickly, however, with the next 3 largest connected sets respectively containing 100, 66, and 55 managers and 67, 44, and 32 stores."

- These can all be found in X_analyze_FE.do, under the block of code:

```
noisily{
display in red "===== Table 1 Column 2 ====="
tab connected_set if connected_set <= 4
}

// How many stores in each connected set?
preserve
bys store_id: gen num_stores = (_n==1)
collapse (sum) num_stores, by(connected_set)
noisily {
display in red "===== Number of connected sets is ====="
count

display in red "===== Number of stores in largest connected sets is ====="
```

```

tab num_stores if connected_set <= 4
}
restore

// How many managers in each connected set?
preserve
bys manager_id: gen num_managers = (_n==1)
collapse (sum) num_managers, by(connected_set)
noisily {
display in red "===== Number of managers in largest connected sets is ====="
tab num_managers if connected_set <= 4
}
restore

```

- **p. 31** “In Company A the tenure gap is 14 months and significantly different from zero at the 1% level. The gap is smaller in Company B, only 3.4 months and not statistically significant at the usual levels.”

- A_event_study.do: `ttest mng_tenure_pos_chg==0` and B_event_study.do: `ttest mng_tenure_start_chg==0`

- **p.52** “In Company A the correlation between store fixed effects under the two methods is 0.98, and the correlation between manager fixed effects is 0.76. In Company B, these correlations are 0.93 and 0.80, respectively”

- These can be found in X_analyze_FE.do as the output of

```

corr manager_fe manager_fe_inter
corr store_fe store_fe_inter

```

- **p. 72** “Averaging these ratios, we find that the mean of r_m^{90} stands at about 0.005, with a standard deviation of 0.112. Increasing the percentile to 99 changes the average ratio r_m^{99} to about 0.021, with a standard deviation of 0.568.”

- Displayed after running A_analyze_FE.do, as the output of

```
noisily display in red "===== Numbers in Appendix E.2 ====="
```

```

sort var_t
gen mean_ratio = mean_v/var_t
noisily su mean_ratio

```

```

gen max_ratio = max_v/var_t
noisily su max_ratio

```

```

gen p90_ratio = p90_v/var_t
noisily su p90_ratio

```

```

gen p99_ratio = p99_v/var_t
noisily su p99_ratio

```

- **p. 73** “about USD 700,000 each month” // “about GBP 4.3 million”

- X_revenue_counterfactuals.do: produced as the output of `bootstrap sg = r(sales_growth) sl = r(sales_level), reps(100) seed($seed_num): assortative_match`

Table 1: Datasets Used in Paper

	Dataset	Origin	Description
Company A	A_dataset.dta	Source data	Dataset obtained from retail companies (cleaned)
	A_connected.dta	Constructed; see file named: A_estimate_fe.R	Identifies all connected sets in data
	A_manager_fe.dta	Constructed; see file named: A_estimate_fe.R	Contains the initial manager fixed effect estimates
	A_store_fe.dta	Constructed; see file named: A_estimate_fe.R	Contains the initial store fixed effect estimates
	A_manager_gid.dta	Constructed; see file named: A_estimate_gfe.R	Contains the manager group id
	A_manager_gfe.dta	Constructed; see file named: A_estimate_gfe.R	Contains the manager grouped fixed effect estimates
	A_store_gid.dta	Constructed; see file named: A_estimate_gfe.R	Contains the store group id
	A_store_gfe.dta	Constructed; see file named: A_estimate_gfe.R	Contains the store grouped fixed effect estimates
	A_dataset_fe_eb.dta	Constructed; see file named: A_analyze_FE.do	Matches the manager and store fixed effects to the source data and computes the EB-adjusted fixed effect estimates
	A_dataset_gfe.dta	Constructed; see file named: A_analyze_GFE.do	Matches the manager and store grouped fixed effects to the source data
	A_event_study.dta	Constructed; see file named: A_event_study.do	Prepares the "A_dataset_fe_eb.dta" to run the event studies
Company B	B_dataset.dta	Source data	Dataset obtained from retail companies (cleaned)
	B_connected.dta	Constructed; see file named: B_estimate_fe.R	Identifies all connected sets in data
	B_manager_fe.dta	Constructed; see file named: B_estimate_fe.R	Contains the initial manager fixed effect estimates
	B_store_fe.dta	Constructed; see file named: B_estimate_fe.R	Contains the initial store fixed effect estimates
	B_manager_gid.dta	Constructed; see file named: B_estimate_gfe.R	Contains the manager group id
	B_manager_gfe.dta	Constructed; see file named: B_estimate_gfe.R	Contains the manager grouped fixed effect estimates
	B_store_gid.dta	Constructed; see file named: B_estimate_gfe.R	Contains the store group id
	B_store_gfe.dta	Constructed; see file named: B_estimate_gfe.R	Contains the store grouped fixed effect estimates
	B_dataset_fe_eb.dta	Constructed; see file named: B_analyze_FE	Matches the manager and store fixed effects to the source data and computes the EB-adjusted fixed effect estimates
	B_dataset_gfe.dta	Constructed; see file named: B_analyze_GFE.do	Matches the manager and store grouped fixed effects to the source data
	B_event_study.dta	Constructed; see file named: B_event_study.do	Prepares the "B_dataset_fe_eb.dta" to run the event studies
NAM Simulations	simulate_match.dta	Constructed; see file named: selection_based_NAM.do	Keeps track of simulated manager-store matches; not used for empirical results

Note: None of the datasets can be made publicly available. See below for a description of the code scripts.

Table 2: Data Dictionary: Variables Contained in "A_dataset"
(Company A)

Variable Name	Description
id	(manager,store) pair id
store_id	store id (numeric)
manager_id	manager id (numeric)
time	period id
year	year of observations
month	month of observation (name)
month_num	month of observation (numeric)
location	name of store's location
location_num	store location (numeric)
state	US State of store (name)
state_num	US State of store (numeric)
county	County of store (name)
countyfip	County FIP
statefip	State FIP
postal_code	postal code
timezone	time zone of store
remodel	indicator if store was remodeled
(comp name)startdate	date that mng started in company
positionstartdate	date that mng started position
job_description	job title
paystatus	pay status
supervisor_id	
supv_title	supervisor title
pos_start_month	month that mng started position
pos_start_year	year that mng started position
(comp name)_start_month	month that mng started in company
(comp name)_start_year	year that mng started in company
manager_tenure_pos	mng tenure in position
manager_tenure_comp	mng tenure in company
sales	total sales in month
labor_cost	cost of labor
manager_salary	manager's salary
pt_count	number of part time employees
ft_count	numner of full-time employees
oh_cost_consigned	overhead cost (consigned)
oh_cost_owned	overhead cost (owned)
oh_cost_total	overhead cost (total)
sales_area	size of sales area in store
store_area	size of store
fte_count	number of full-time equivalent employees
log_sales	log of sales
log_manager_salary	log of manager salary
log_manager_tenure_pos	log of tenure in position
log_manager_tenure_comp	log of tenure in company
log_ft_count	log of full-time employees
log_pt_count	log of part-time employees
log_fte_count	log of full-time equivalent employees
log_store_area	log of store area

log_sales_area	log of sales area
capital_cost	Cost of capital
oh_cost	overhead cost (same as oh_cost_total)
log_oh_cost	log of overhead cost
closest_competitor	distance to closest competitor (miles)
closest_[competitor 1]	distance to closest competitor 1 (competitor's name in brackets)
closest_[competitor 2]	distance to closest competitor 2
closest_[competitor 3]	distance to closest competitor 3
closest_[competitor 4]	distance to closest competitor 4
num_mng_2019	number of different managers in 2019
num_close_competitors	number of close competitors (within 1 mile)
sun_open_hour	hour that store opens on Sunday
sun_close_hour	hour that store closes on Sunday
sun_num_hours	number of hours open on Sunday
mon_open_hour	hour that store opens on Monday
mon_close_hour	similar as above
mon_num_hours	
tue_open_hour	
tue_close_hour	
tue_num_hours	
wed_open_hour	
wed_close_hour	
wed_num_hours	
thu_open_hour	
thu_close_hour	
thu_num_hours	
fri_open_hour	
fri_close_hour	
fri_num_hours	
sat_open_hour	
sat_close_hour	
sat_num_hours	
week_num_hours	number of hours open during week days
weekend_num_hours	number of hours open during weekends
sun_log_hours	log of hours open on Sunday
mon_log_hours	log oh hours open on Monday
tue_log_hours	similar to the above
wed_log_hours	
thu_log_hours	
fri_log_hours	
sat_log_hours	
week_log_hours	
weekend_log_hours	

Table 3: Data Dictionary: Variables Contained in "B_dataset"
(Company B)

Variable Name	Description
store_id	store id (numeric)
manager_id	manager id (numeric)
time	time period
year	year of observation
month	month of observation
elec_consumption	consumption of electricity in month
elec_cost	cost of electricity in month
gas_consumption	consumption of gas in month
gas_cost	cost of gas in month
competitor_count	number of close competitors
mean_dist_km	mean distance to competitors in kilometers
mean_drivetime	mean drivetime to competitors
mean_dist_drive_km	mean driving distance to competitors in kilometers
min_distance_km	minimum distance to competitor in kilometers
min_drivetime	minimum drive time to competitor
min_dist_drive_km	minimum driving distance to competitor in kilometers
[competitor 1]	distance to [competitor 1] (name of competitor in brackets)
[competitor 2]	distance to [competitor 2]
[competitor 3]	distance to [competitor 3]
[competitor 4]	distance to [competitor 4]
[competitor 5]	distance to [competitor 5]
[competitor 6]	distance to [competitor 6]
[competitor 7]	distance to [competitor 7]
area_total	size of store
area_sales	size of sales area in store
store_name	store's name
format	store format (flagship, local, others)
format_num	store format (numeric)
type	store type (convenience, large, small, others)
opening_date	store opening date
region	store region (numeric)
region_name	store region name
area	store area (location, numeric)
area_name	store area name
town	store town
county	store county
postcode	store postal code
longitude	store's location longitude
latitude	store's location latitude
country	store country
location_code	store's location code
saturday_opening	Saturday opening hour
saturday_closing	Saturday closing hour
sunday_opening	similar to above
sunday_closing	
weekday_opening	
weekday_closing	
weekday_hours	number of minutes open during weekdays

saturday_hours	number of minutes open during Saturdays
sunday_hours	number of minutes open during Sundays
weekday_hours_total	total number of minutes open during weekdays
weekend_hours	number of minutes open during the full weekend
week_hours	total number of minutes open over the entire week (including weekends)
revenue_total	total revenue during month
manager_startdate	date when manager started job
months_on_job	number of months that manager has spent on job
hc_count	number of employees
fte_count	number of full-time equivalent employees
elec_use_sqft	electricity consumption per squared foot in store
gas_use_sqft	gas consumption per squared foot in store
elec_price	electricity price (constructed as cost/consumption)
gas_price	gas price (constructed as cost/consumption)
energy_cost	energy (electricity + gas) cost
log_revenue_total	log of total revenue
log_energy_cost	log of energy cost
log_fte_count	log of full-time equivalent employees
log_hc_count	log of number of employees
location_id	store's city/location (numeric)
location_name	store's city/location name
location_type	location type (admin county, metropolitan, borough, etc.)
number_active_managers	number of active manager in store
multiple_managers	indicator of multiple managers in store
number_stores	number of store that manager in observation works in
multiple_stores	indicator of manager that works in multiple stores
open_year	year when store was opened
open_month	month when store was opened
store_age	store's age
log_store_age	log of store age
log_store_area	log of store size
log_week_hours	log of minutes open during weekdays
log_weekend_hours	log of minutes open during weekends
female	indicator of female manager
title	manager title (Mr., Mrs., Ms., others)
firstname	manager's first name
lastname	manager's last name
doj	date when manager joined company
join_year	year when manager joined company
join_month	month when manager joined company
manager_tenure	manager's tenure in job/position
manager_tenure_start	manager's tenure in company
log_manager_tenure	log of manager tenure in position
log_manager_tenure_start	log of manager tenure in company when they became a manager
